

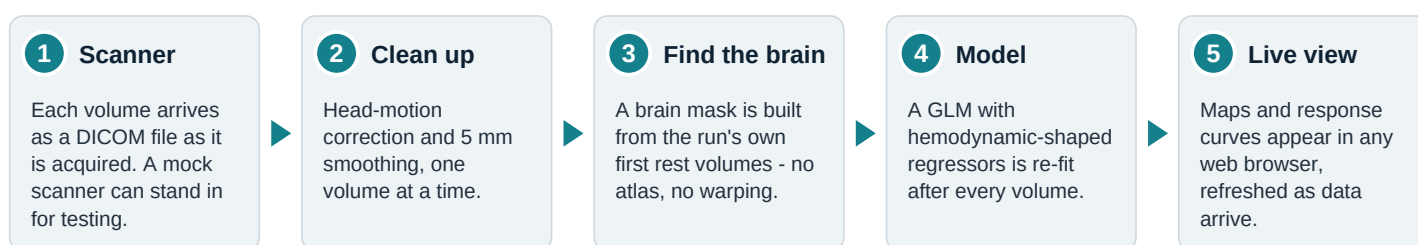
# Real-Time fMRI Task Activation

Watch the brain respond while the scan is still running - no anatomical scan needed.

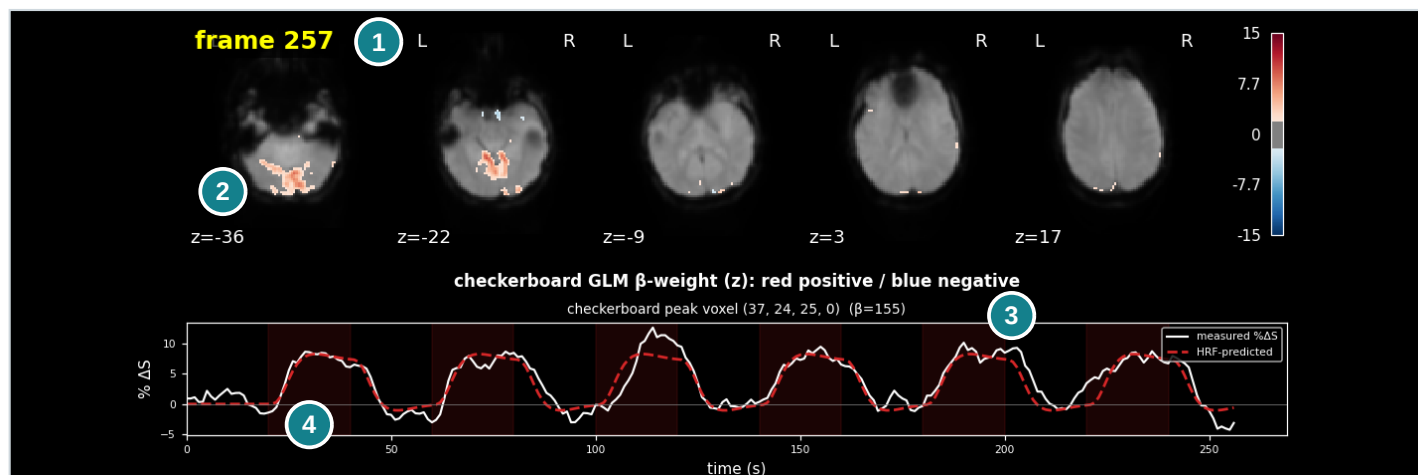
## WHAT IT DOES

As each fMRI volume comes off the scanner it is corrected for head motion, lightly smoothed, and passed to a statistical model that is **updated after every volume**. A web page shows where the brain is responding and how closely each region follows the task - as early as the first task block, while the participant is still in the scanner. Nothing has to be prepared in advance: **no anatomical scan, atlas or registration** is needed, and any block or event design can be loaded from a standard events file.

## HOW IT WORKS



## READING A LIVE FRAME



Sample live output: the visual (checkerboard) task, near the end of the run.

- 1 Frame counter** - which volume is on screen.
- 2 Activation map** - voxels that respond to the task, drawn over the participant's own brain image. Red = increase, blue = decrease.
- 3 Measured vs. predicted** - white: the measured signal (% change) at the most task-responsive voxel; dashed: the textbook hemodynamic response to this design.
- 4 Task timing** - shaded bands mark when the stimulus was on. A good response rises and falls with them.

**Why "registration-free"?** Because the brain mask and regions of interest come from the participant's own functional images, the demo starts working right away and needs no anatomical scan or template alignment. The trade-off: results are shown in the participant's native space (slice heights are scanner millimetres, not standard MNI coordinates), so treat anatomy labels as approximate.

# Example fMRI Demonstration Tasks

Each task ships with a stimulus script and a matching analysis config; the strips show live-view slices.

## Visual Perception Task

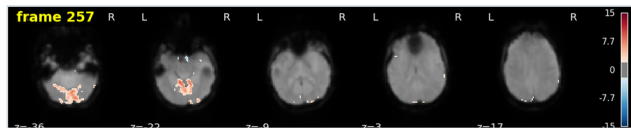
Checkerboard vs rest | 260 s

VISION

A flickering checkerboard alternates with a fixation cross. Eyes stay on the center of the screen the whole time.



**Expect:** Visual cortex at the back of the brain responds while the checkerboard is on.

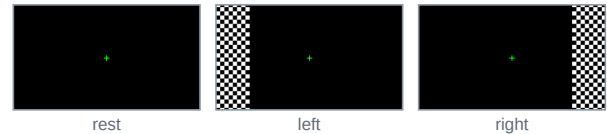


## Visual Perception Task

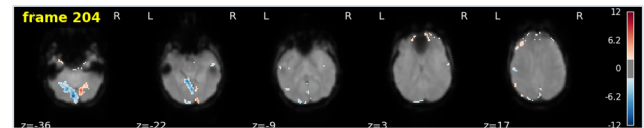
Left vs right visual field | 204 s

VISION

Checkerboard bars flash at the left or right edge of the screen while the eyes hold the center cross.



**Expect:** Each half of the visual field drives the **opposite** hemisphere.



## Voluntary Movement Task

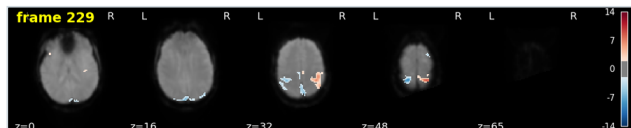
Left vs right hand | 250 s

MOVEMENT

Squeeze the cued hand into a fist, like a stress ball, then relax - again and again while the cue stays up. Relax on the + cross.



**Expect:** The **opposite** side of motor cortex lights up (left hand = right hemisphere).



## Hand Guessing Game

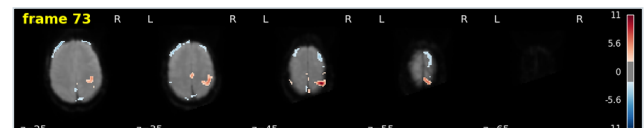
Secret hand | 250 s

GAME

The participant **secretly** picks a hand and moves it whenever MOVE shows. Nobody tells the experimenter - the group guesses from the live map.



**Expect:** Activation sits on one side of motor cortex; the opposite hemisphere gives the answer away.



## Blackjack Game

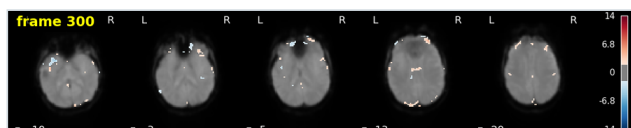
Win vs lose | 58 trials, about 5 min

REWARD

Two cards, then hit or stay. A pre-scripted win, loss or tie is revealed (24 wins, 24 losses, 10 ties).



**Expect:** Reward-related regions such as the striatum; these effects are typically subtler than motor or visual ones.

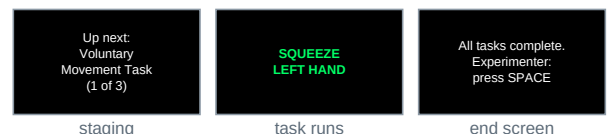


## Run several back-to-back

Psychtoolbox battery | one shared window

RUNNER

A battery runner opens **one** presentation window for the whole session, so the screen never drops back to the desktop between tasks.



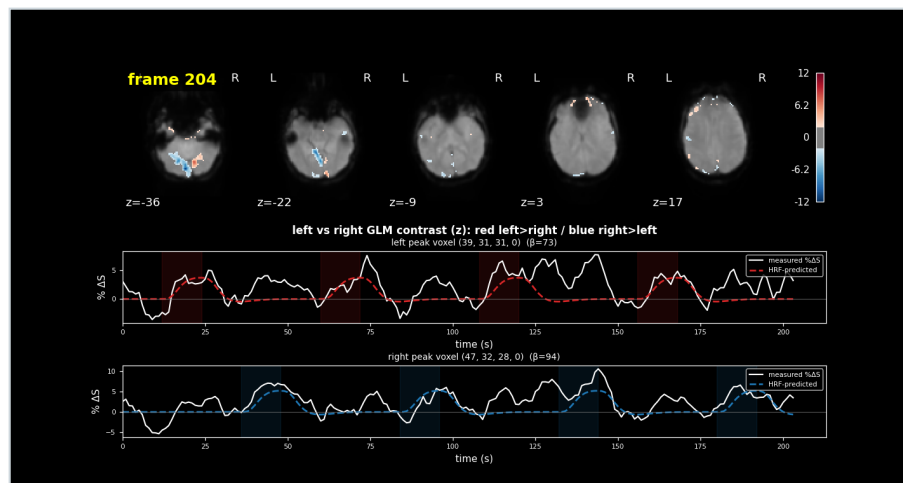
**Between tasks:** an "Up next" screen holds until the experimenter presses SPACE (Escape stops the session cleanly).

**Friendly names:** participants see "Visual Perception Task", "Voluntary Movement Task", "Blackjack Game" or "Hand Guessing Game" - never the file names.

**How to:** see the run\_battery code on page 3.

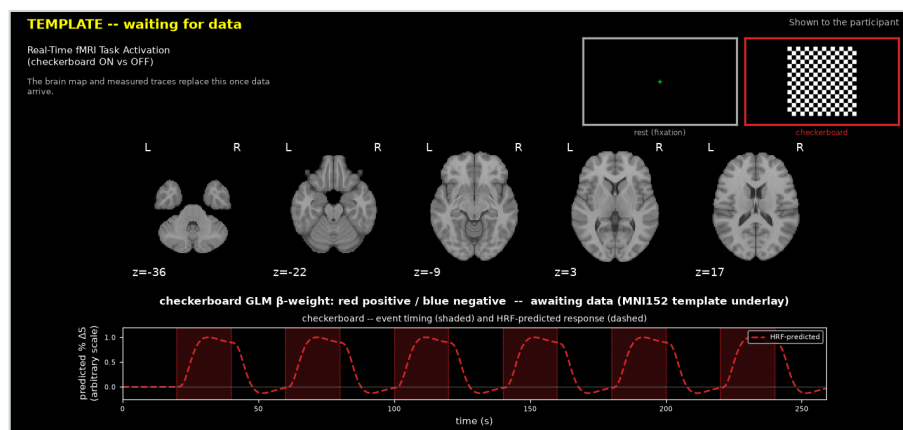
# Sample output, and how to run it

A live-view frame, the pre-data template, then how to run the analysis and the stimuli.



## Visual task: left vs right

End of a full run (frame 204). Red marks voxels that respond more to the **left** stimulus, blue more to the **right**. Each lower row compares the measured signal (white) with the modelled response (dashed) at that condition's peak voxel, with the stimulus blocks shaded.



## Before the first volume arrives

Each task starts with a template so the screen is never empty: the slices it will show, the screens the participant sees (top right), and empty response curves with the expected timing shaded. The first live frame replaces it (shown here for the checkerboard task).

## TRY IT YOURSELF

### START THE ANALYSIS (ONE COMMAND PER TASK)

```
./quickstart.sh checkerboard_1cond --run 7
./quickstart.sh checkerboard_2cond --run 9
./quickstart.sh motor --run 11
```

--run N is optional: it sets the scan run number (7, 9, 11...).

Then open `outDir/live/viewer.html` in a browser.

### You will need

- Docker (the `brainiak/rtcloud` image, which includes FSL)
- A scanner's DICOM stream - or the included mock scanner for a no-scanner test
- PsychoPy or MATLAB + Psychtoolbox on the stimulus computer
- A web browser to watch the live view

### Good to know

- A research demo - not a clinical tool. Maps are thresholded and shown in native space.

## RUN THE STIMULI WITH RUN\_BATTERY (PSYCHTOOLBOX)

### IN MATLAB, ON THE STIMULUS COMPUTER

```
cd stimuli_ptb
run_battery({ ...
    {@checkerboard_1cond_task, {}}, ...
    {@checkerboard_2cond_task, {}}, ...
    {@motor_task, {}}, ...
})
```

- **One shared window** stays open for the whole session - no flashing back to the MATLAB desktop between tasks.
- Before each task an **"Up next"** screen appears; the experimenter presses SPACE to begin it (Escape stops the battery). An "All tasks complete" screen closes the session.
- List tasks in the same order as the commands above so each staging screen pairs with its analysis run (7, 9, 11).
- For a desk test, add `'TriggerKey', {'space'}` to a task and `'Windowed', true` to the call.